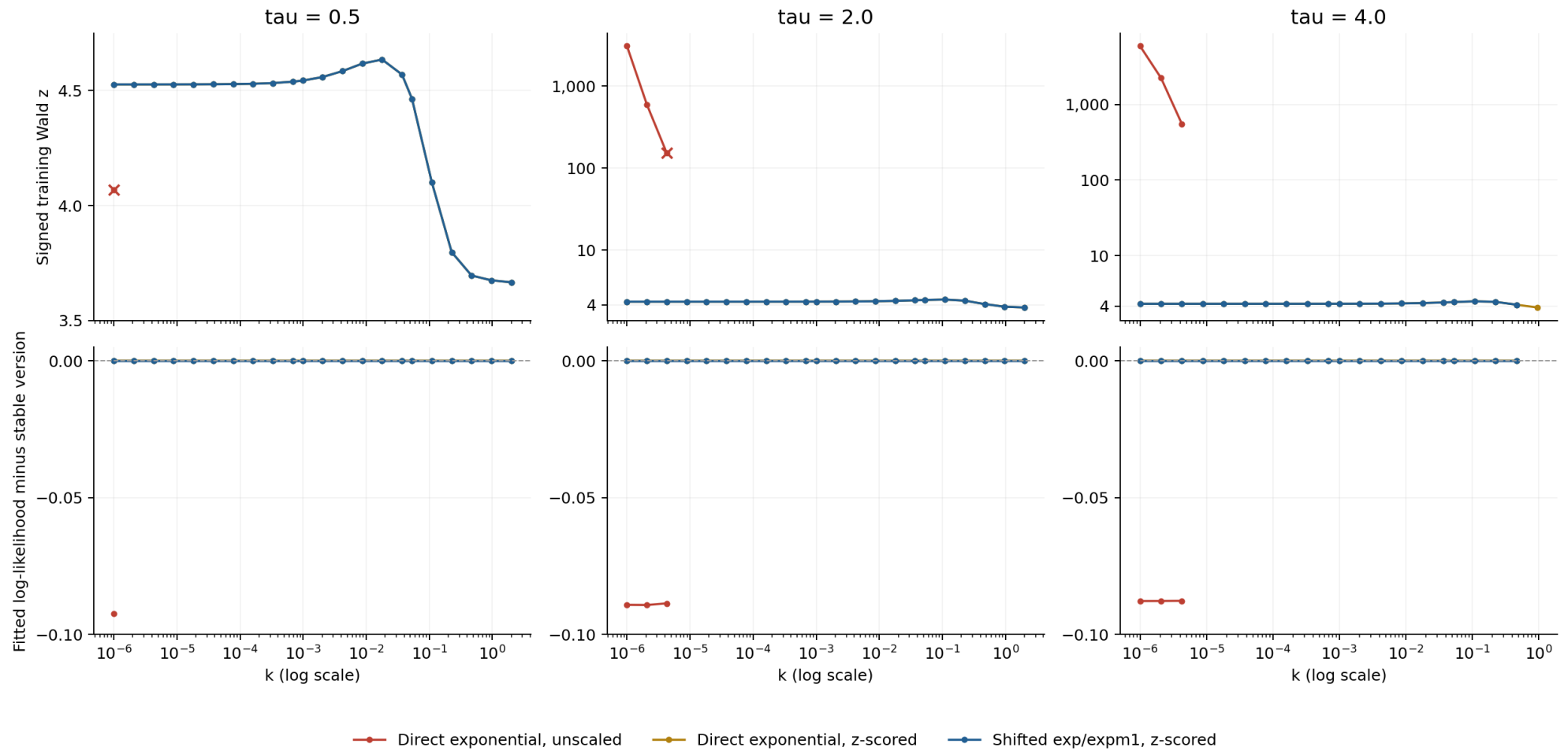


Numerical check of the X21 exponential predictor



Same Tr-24 training responses and GLMM. Upper panels: linear scale at $\tau=0.5$; symmetric-log above $z=10$ otherwise.
 Limited to four minutes; missing points were not completed. Crosses flag nonconverged or singular fits.

This four-minute check saved 136 of 198 planned fits: 7 unscaled, 65 directly exponentiated and z-scored, and 64 stabilized and z-scored. All 7 saved unscaled fits had warnings; 2 also failed convergence checks. Incomplete batches were stopped at the time limit. No missing result is treated as a zero.

What was compared

I reused the X21 HuBERT-base Tr-24 3-D t-SNE distances from the completed parameter map: 10,969 word responses from training folds 1 and 2, with all conditions retained. I compared tau = 0.5, 2 and 4, at 22 positive k values per slice, using identical responses and the same binomial-logit GLMM. Tau = 0.5 is a non-metric sensitivity case.

Each candidate used lme4 1.1-35.5, bobyqa, nAGQ = 1 and maxfun = 20,000. The fixed terms were an intercept, the candidate predictor and test-talker identity; random intercepts were included for participant and analysis item. Experimental condition was not included. No feature extraction, t-SNE or DTW calculation was rerun.

```
raw <- exp(-k * d$raw_distance)
a <- -k * (d$raw_distance - min(d$raw_distance))
stable <- if (max(abs(a)) < 0.1) expm1(a) else exp(a)
# Three separate fits: raw; (raw-mean(raw))/sd(raw); (stable-mean(stable))/sd(stable)
```

At tau = 2, k = 0.000001	Unscaled exp	Stable + z-score
Wald z	3065.8409	4.3336
Fitted log likelihood	-3893.3028	-3893.2137
Predictor-scale warning	Yes	No

At this parameter pair the raw predictor SD was about 0.00000604. This almost-constant, unscaled predictor produces a numerically unreliable Wald statistic. The raw fit reported an optimizer convergence code of zero, illustrating why a convergence code alone is insufficient when scale warnings remain.

Across matched completed points, direct exp followed by z-scoring and the stable implementation differed in z by at most 0.0000368. Z-scoring is therefore the main effective correction demonstrated here. The shifted exponential and expm1 additionally protect against common-factor underflow and small-value cancellation. They are affine transformations before standardization, not smoothing of the fitted surface.

Scope: this is a time-limited numerical check on one training split, not a reconstruction of the historical screenshot, a complete raw-predictor surface, or a held-out performance comparison. The historical irregularities may have additional causes. Exact fit outputs, warnings, source hashes and scripts are retained in [outputs/florian_email_2026-09-21/](#).